

TRAINING RECOMMENDATION SYSTEM

PROJECT REPORT

Submitted by

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*In partial fulfilment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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APRIL 2026

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DECLARATION

I affirm that the project work titled “**Training Recommendation System**” being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology in Artificial Intelligence and Data Science** is the record of original work done by us under the guidance of **Mr. Navaneeth Kumar K**, Assistant Professor Level III, Department of Artificial Intelligence and Data Science. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

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ACKNOWLEDGEMENT

I would like to enunciate heartfelt thanks to our esteemed Chairman **Dr.S.V. Balasubramaniam**, and the respected Principal **Dr. C. Palanisamy** for providing excellent facilities and support during the course of study in this institute.

I are grateful to **Dr. Gomathi R, Head of the Department, Department of Artificial Intelligence and Data Science** for her valuable suggestions to carry out the project work successfully.

I wish to express our sincere thanks to Faculty guide **Mr. Navaneeth Kumar K, Assistant Professor Level III, Department of Artificial Intelligence and Data Science**, for his constructive ideas, inspirations, encouragement, excellent guidance, and much needed technical support extended to complete our project work.

I would like to thank our friends, faculty and non-teaching staff who have directly and indirectly contributed to the success of this project.

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ABSTRACT

An Intelligent Training Recommendation System is a web-based application designed to provide personalized learning guidance by analyzing user performance, interests, and skill levels. The system integrates a user-friendly frontend with a Python Flask backend and a structured database to efficiently manage user data, assessments, and recommendations. It aims to enhance the learning experience by guiding students toward suitable training resources and improving their overall skill development.

The system supports multiple user roles, including students and trainers. Students can register, log in, attempt MCQ-based assessments, upload content for OCR-based analysis, and receive tailored training recommendations based on their performance. The system evaluates user responses and identifies strengths and weaknesses to suggest appropriate learning paths. Trainers can monitor student progress, analyze results, and provide guidance through a dedicated dashboard.

Key features of the system include automated MCQ evaluation, OCR-based text extraction for analysis, personalized recommendation generation, and real-time performance tracking. The use of Flask ensures efficient backend processing, while the modular design allows scalability and easy maintenance. The system improves decision-making in learning by combining assessment data with intelligent recommendation techniques.

Overall, this project demonstrates how web technologies and intelligent systems can be integrated to create a scalable and efficient training platform, promoting personalized learning and better academic outcomes.

Keywords : Training Recommendation System, Flask, Python, Machine Learning, OCR, MCQ Assessment, Personalized Learning, Web Application, Student Dashboard, Trainer Dashboard, Performance Analysis, Recommendation Engine, Skill Development, Educational Technology (EdTech), Full Stack Development, Data Processing, User Authentication

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CHAPTER - I

INTRODUCTION

The Intelligent Training Recommendation System is a web-based application designed to enhance personalized learning by analyzing user performance and providing suitable training suggestions. In traditional learning environments, students often struggle to identify the right resources or courses that match their skill levels. This leads to inefficient learning paths, lack of direction, and reduced productivity. The proposed system addresses these challenges by offering an automated and intelligent platform that guides users toward appropriate learning opportunities.

The system is developed using Python with the Flask framework for backend processing, along with HTML, CSS, and JavaScript for creating an interactive and user-friendly frontend. A structured database is used to store user data, test results, and recommendations efficiently. These technologies work together to ensure smooth performance, fast response, and scalability of the application.

The application mainly supports two user roles: Students and Trainers. Students can register, log in, attempt MCQ-based assessments, and upload images or documents for OCR-based text extraction. Based on their performance and extracted data, the system analyzes their strengths and weaknesses to generate personalized training recommendations. Trainers can monitor student progress, review performance data, and guide learners through a dedicated dashboard.

The system integrates key features such as automated MCQ evaluation, Optical Character Recognition (OCR) for extracting textual data, and a recommendation engine that suggests relevant training content. Role-based access ensures secure and organized usage of the platform by restricting functionalities based on user roles.

The main objective of this system is to reduce manual effort in analyzing student performance and to improve learning outcomes through intelligent recommendations. By

combining assessment, analysis, and suggestion in a single platform, the system enhances efficiency, decision-making, and overall learning experience. Thus, the Intelligent Training Recommendation System provides a scalable and effective solution for modern educational needs.

1.1 BACKGROUND OF THE STUDY

In recent years, the rapid growth of digital learning platforms has transformed the way education is delivered and accessed. With the increasing availability of online courses, training programs, and educational resources, learners are exposed to a vast amount of information. However, this abundance often creates confusion, as students find it difficult to identify the most relevant learning materials that match their individual skill levels and interests. Traditional learning systems lack personalization and do not provide proper guidance for skill development.

In many educational environments, student performance evaluation is limited to manual assessments, which do not effectively analyze strengths and weaknesses. Additionally, there is a lack of systems that can integrate multiple functionalities such as assessment, analysis, and recommendation into a single platform. As a result, students may follow inefficient learning paths, leading to reduced motivation and productivity.

With advancements in technologies such as Machine Learning, Web Development, and Optical Character Recognition (OCR), it has become possible to design intelligent systems that can analyze user data and provide meaningful insights. Recommendation systems, which are widely used in platforms like e-commerce and entertainment, can also be applied in education to suggest personalized learning resources based on user behavior and performance.

The need for an intelligent training recommendation system arises from these challenges. By combining features such as MCQ-based assessments, OCR-based data extraction, and performance analysis, the system can better understand user capabilities and provide targeted recommendations. This not only improves learning efficiency but also helps in skill enhancement and career development

1.2 SCOPE OF THE PROPOSED WORK

The scope of the proposed Intelligent Training Recommendation System focuses on developing a web-based platform that provides personalized learning guidance to users based on their performance and skill level. The system is designed to integrate multiple functionalities such as user authentication, assessment, data analysis, and recommendation into a single platform, making the learning process more efficient and structured.

The system allows users to register and log in securely, ensuring proper access control. Once authenticated, students can take MCQ-based assessments to evaluate their knowledge in different domains.

In addition to this, the system supports OCR (Optical Character Recognition) functionality, which enables users to upload images or documents, from which text is extracted and analyzed for further processing.

Based on the assessment results and extracted data, the system performs performance analysis to identify strengths and weaknesses of the user. Using this analysis, the recommendation engine suggests suitable training materials, courses, or skill improvement areas tailored to individual needs. This helps students follow a more effective and personalized learning path.

The system also includes role-based dashboards for students and trainers. Students can track their progress, view recommendations, and improve their skills, while trainers can monitor student performance and provide guidance accordingly. The modular design ensures scalability, allowing future enhancements such as advanced machine learning models and integration with external learning platforms.

CHAPTER – II

LITERATURE SURVEY

Personalized training recommendation systems have gained significant attention in educational research due to their potential to improve learning efficiency and engagement. Smith (2018) highlighted the limitations of traditional learning environments, where students often struggle to select training resources that match their skill levels, leading to inefficient learning paths and reduced productivity.

To address these issues, automated systems that analyze learner performance and provide tailored recommendations were proposed. Clark (2019) demonstrated that integrating automated assessment and feedback mechanisms reduces the manual workload for educators and enhances the learning experience by providing timely, performance-based guidance.

Similarly, Johnson (2019) and Williams (2020) explored web-based and cloud-integrated platforms, showing that centralized, accessible systems allow learners to receive training recommendations in real time, improving both engagement and scalability.

Algorithmic approaches have been central to the development of effective recommendation systems. Lewis (2020) studied content-based and collaborative filtering methods, emphasizing that hybrid models combining both approaches increase accuracy by considering user performance as well as content similarity.

Walker (2020) highlighted the benefits of integrating intelligent algorithms with user assessment data, showing that systems can identify individual strengths and weaknesses and generate focused learning paths. Security and usability have also been emphasized in literature.

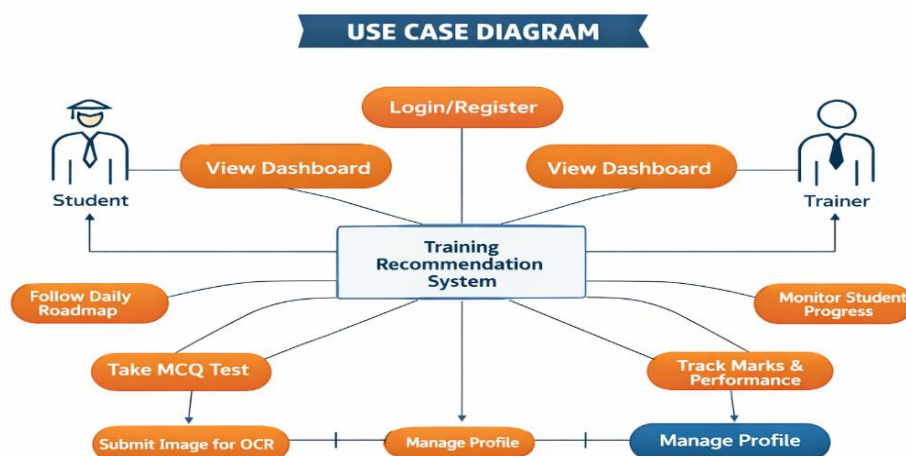
Brown (2021) demonstrated the importance of role-based access control (RBAC) for securing sensitive user data, while Davis (2021) noted that intuitive interfaces and smooth navigation are critical for encouraging system adoption among both technical and non-technical users.

Recent studies focus on advanced technological integration for enhanced system efficiency. Miller (2022) and Anderson (2022) highlighted the role of modern web frameworks such as FLASK in creating responsive, dynamic, and scalable platforms capable of handling multiple users simultaneously.

Thomas (2023) showed that databases like MongoDB or relational alternatives support flexible schema designs, enabling storage of dynamic datasets including assessment results, user profiles, and recommendation histories.

The use of AI techniques, OCR-based text extraction, and automated evaluation has been emphasized by White (2023) and Harris (2024), showing that these technologies enable systems to analyze performance and provide tailored recommendations without manual intervention.

Lee (2024) concluded that integrating modern backend and frontend frameworks with intelligent recommendation engines results in scalable, efficient, and user-friendly training systems capable of enhancing personalized learning outcomes.



2.1.1 Use Case Diagram

CHAPTER - III

OBJECTIVES AND METHODOLOGY

3.1 OBJECTIVE

The primary objective of the Training Recommendation System is to design and develop an intelligent, user-friendly, and secure web-based platform that provides personalized learning guidance for students while enabling trainers to effectively monitor and evaluate student performance. The system aims to bridge the gap between traditional learning methods and modern digital education by integrating automation, personalization, and data-driven insights.

One of the key objectives of this system is to provide personalized learning recommendations. Every student has different learning capabilities, goals, and progress levels. The system analyses student inputs such as current course, course level, and learning goals to generate a customized daily roadmap. This roadmap acts as a structured plan that guides students step-by-step, helping them stay organized, focused, and consistent in their learning journey. By tailoring content to individual needs, the system enhances learning efficiency and improves outcomes.

Another important objective is to ensure secure and reliable user authentication. The system incorporates a robust login and registration mechanism using password hashing and JSON Web Token (JWT) authentication. Password hashing ensures that sensitive user credentials are stored securely, reducing the risk of data breaches. JWT authentication enables safe communication between the client and server, ensuring that only authorized users can access their respective dashboards. This strengthens the overall security of the platform and builds user trust.

The system also aims to simplify and automate the learning process. Traditional learning methods often lack structured guidance and require manual effort to track progress. This system automates various tasks such as roadmap generation, progress tracking, and performance evaluation. Students no longer need to manually plan their study schedules, as

the system provides a clear and dynamic roadmap. This reduces confusion and saves time, allowing students to focus more on learning rather than planning.

An additional objective is to enhance learning through technology integration, particularly with the inclusion of an Optical Character Recognition (OCR) module. This feature allows students to upload images containing textual information, extract the text, and display key points in a readable format. This is especially useful for digitizing handwritten notes or printed materials. By converting visual data into editable text, the system improves accessibility and helps students better understand and revise content.

The system further aims to support self-assessment and continuous evaluation. The inclusion of a Multiple Choice Questions (MCQ) test module enables students to test their knowledge and evaluate their understanding of various topics. Immediate feedback and results help students identify their strengths and weaknesses. This continuous self-evaluation encourages active learning and helps students improve over time.

Another major objective is to facilitate effective monitoring and evaluation by trainers. The trainer dashboard provides tools to track student progress, analyze performance, and monitor test results. Trainers can easily identify students who are excelling and those who need additional support. This enables better guidance, timely intervention, and improved teaching strategies. The system thus creates a collaborative learning environment between students and trainers.

The system also focuses on efficient data management and organization. All user information, course details, test results, and progress data are stored in a structured MySQL database. This ensures data consistency, easy retrieval, and scalability. Proper database management allows the system to handle multiple users simultaneously without performance issues.

Another objective is to provide a user-friendly and interactive interface. The frontend of the system is developed using HTML, CSS, and JavaScript to ensure a responsive and visually appealing design. The interface is designed to be intuitive, allowing users to navigate through different features easily. A well-designed UI enhances user experience and encourages regular usage of the platform.

The system also aims to ensure seamless integration and communication between components. The backend is developed using Python Flask, which handles API routing and communication between the frontend and database. All functionalities, such as login, roadmap generation, OCR processing, and test evaluation, are integrated through APIs. These APIs are tested using Postman to ensure reliability, accuracy, and smooth performance.

Finally, the Training Recommendation System aims to promote smart learning and digital transformation in education. By combining personalized recommendations, automated processes, secure authentication, and advanced features like OCR and self-evaluation, the system provides a modern solution to learning challenges. It encourages students to adopt technology-driven learning methods and supports trainers in delivering better educational outcomes.

3.1.1 PURPOSE OF THE PROJECT

The purpose of the Training Recommendation System is to develop an efficient, secure, and user-friendly web-based platform that enhances the learning experience of students while enabling trainers to effectively monitor and guide their progress. The system is designed to address the limitations of traditional learning methods by introducing automation, personalization, and intelligent recommendations.

In conventional learning environments, students often face difficulties in planning their study schedules, identifying the right learning resources, and tracking their progress. There is usually a lack of structured guidance, which can lead to confusion, inconsistency, and reduced productivity. The primary purpose of this project is to overcome these challenges by providing a system that generates a personalized learning roadmap based on each student's current course, level, and goals. This helps students follow a clear and organized path toward achieving their objectives.

Another important purpose of the system is to ensure secure and reliable user management. By implementing password hashing and JWT-based authentication, the project ensures that

user data is protected and access to the system is restricted to authorized individuals only. This creates a safe and trustworthy environment for both students and trainers.

The project also aims to integrate advanced technologies to improve learning efficiency. The inclusion of an Optical Character Recognition (OCR) module allows users to upload images and extract textual information, making it easier to convert handwritten or printed notes into digital format. This enhances accessibility and supports better understanding and revision of study materials.

Additionally, the system is designed to promote self-learning and continuous evaluation. The MCQ-based test module enables students to assess their knowledge and receive immediate feedback on their performance. This helps them identify their strengths and weaknesses and take necessary steps to improve.

From the trainer's perspective, the purpose of the project is to provide a centralized platform for monitoring student performance and progress. Trainers can track student activities, evaluate test results, and analyze overall performance efficiently. This enables better decision-making and personalized guidance for each student.

Another key purpose is to ensure efficient data management and seamless system operation. The use of MySQL for database management ensures organized storage and quick retrieval of user data, while Python Flask enables smooth communication between the frontend and backend through APIs. Testing with Postman ensures that all functionalities work correctly and reliably.

Finally, the project aims to contribute to the digital transformation of education by providing a smart, interactive, and scalable learning platform. It encourages the adoption of modern technologies in education and supports both students and trainers in achieving better learning outcomes.

3.1.2 TARGET AUDIENCE AND BENEFICIARIES

TARGET AUDIENCE

The Intelligent Training Recommendation System is designed for both students and educators who seek a more personalized, efficient, and interactive learning experience. The primary audience includes students who want to improve their skills, identify areas of weakness, and receive tailored training recommendations based on their performance in assessments or uploaded study materials.

The system is particularly beneficial for learners who may struggle with selecting appropriate resources in traditional educational settings, as it guides them toward optimized learning paths.

Additionally, the system targets trainers and educators who aim to monitor student progress, evaluate performance, and provide timely feedback without the burden of manual tracking and analysis.

Educational institutions and training centers can also benefit, as the platform offers scalable and centralized management of assessments, recommendations, and learner data, ensuring streamlined workflows.

Overall, the system is intended for anyone involved in the learning process—students, trainers, and institutions—who seeks a modern, data-driven, and intelligent solution to enhance teaching and learning outcomes.

BENEFICIARIES

The Intelligent Training Recommendation System provides benefits to multiple groups involved in the learning and training process. The primary beneficiaries are students, who gain personalized guidance to identify their strengths and weaknesses, access appropriate training resources, and follow optimized learning paths, thereby improving knowledge retention and academic performance.

Trainers and educators also benefit, as the system automates assessment evaluation, tracks student progress, and provides insights through a dedicated dashboard, reducing manual effort and enabling more effective mentoring.

Educational institutions and training centers gain from the centralized and scalable management of assessments, learner data, and recommendations, allowing them to implement structured learning programs and improve overall educational efficiency.

Additionally, the system indirectly benefits parents or guardians, who can monitor student progress and support learning at home based on insights provided by the platform. By addressing the needs of these stakeholders, the system fosters a collaborative, efficient, and intelligent learning environment that enhances outcomes for all parties involved.

3.1.3 WIDER IMPACT AND SOCIETAL BENEFITS

The Training Recommendation System has a profound wider impact on society by promoting a more accessible, personalized, and technology-driven approach to education. In the current era, where digital transformation is rapidly reshaping various sectors, education remains one of the most critical areas that require innovation and improvement. Traditional learning systems often follow a one-size-fits-all approach, which may not effectively address the diverse needs, abilities, and learning speeds of individual students. This system addresses that gap by offering personalized learning experiences through dynamically generated roadmaps tailored to each student's goals, course level, and progress. By doing so, it supports inclusive education and ensures that learners receive guidance that suits their individual requirements.

One of the major societal benefits of this system is its ability to enhance accessibility to quality education. Students from rural or underserved areas often lack access to structured learning resources and experienced mentors. This system provides them with a digital platform where they can access organized study plans, learning materials, and self-assessment tools without geographical limitations. As a result, it helps reduce educational

inequality and creates opportunities for students who may otherwise be disadvantaged. The inclusion of features like OCR further supports accessibility by allowing users to convert handwritten or printed materials into digital text, making learning more convenient and efficient.

The system also contributes to the development of self-directed and lifelong learning habits. By providing daily roadmaps and self-evaluation modules such as MCQ tests, students are encouraged to take responsibility for their own learning. They can track their progress, identify areas of improvement, and continuously enhance their knowledge. This fosters independence, discipline, and critical thinking skills, which are essential not only for academic success but also for personal and professional growth. In a rapidly evolving job market, where continuous skill development is necessary, such systems play a crucial role in preparing individuals for future challenges.

Another important impact of the system is its role in improving the efficiency of educators and training processes. Trainers can easily monitor student performance, analyze test results, and identify learning patterns through a centralized dashboard. This reduces the need for manual tracking and allows educators to focus more on mentoring and providing personalized support. With better insights into student progress, trainers can adopt more effective teaching strategies and ensure that no student is left behind. This ultimately leads to improved learning outcomes and a more efficient education system.

3.2 METHODOLOGY

3.2.1 SYSTEM DESIGN AND ARCHITECTURE

SYSTEM ARCHITECTURE

System architecture defines the overall structure of a system and shows how different components interact to perform specific tasks. The architecture of the Training Recommendation System is designed to ensure efficiency, scalability, and easy maintenance. It follows a structured approach where the system is divided into three main layers, each responsible for specific functionalities.

Presentation Layer (User Interface): This layer handles all interactions with users. It provides an interface for students to register, log in, attempt MCQ assessments, upload images or documents for OCR-based text extraction, and view personalized training recommendations. Trainers can also use this layer to monitor student progress and performance through a dashboard. The interface is developed using HTML, CSS, and JavaScript, focusing on simplicity, responsiveness, and ease of navigation.

Application Layer (Business Logic): This layer manages the core functionality of the system. It processes user inputs, evaluates MCQs automatically, extracts text using OCR, and generates personalized training recommendations. Developed using Python with Flask, it also handles communication between the frontend and the database while enforcing role-based access control to ensure secure and organized system usage.

Data Layer (Database): This layer is responsible for storing and managing all system data. Using MySQL, it maintains user profiles, assessment results, OCR-extracted data, and recommendation history in a structured format. The data layer ensures efficient retrieval, data integrity, and secure storage, allowing trainers and students to access information quickly and reliably.

SYSTEM DESIGN

The System Design of the Training Recommendation System focuses on defining the structure, components, and workflows required to implement the system effectively. It provides a detailed blueprint of how different modules interact and how data flows within the system.

The system follows a modular design approach, where each functionality is divided into separate modules to improve maintainability and scalability. The key modules of the system include:

1. User Authentication Module:

This module handles user registration and login functionalities. It verifies user credentials,

encrypts passwords using hashing techniques, and generates JWT tokens for secure session management. It also manages role-based access for students and trainers.

2. Student Management Module:

This module allows students to view and manage their profiles, including personal details, course information, and learning goals. It acts as the central point for student-related activities.

3. Roadmap Generation Module:

This is one of the core components of the system. Based on the student's input (course, level, and goals), the system dynamically generates a daily learning roadmap. This helps students follow a structured path and achieve their objectives efficiently.

4. OCR Module:

The OCR module enables users to upload images and extract text from them. The extracted content is processed and displayed in a structured format, improving readability and usability of study materials.

5. MCQ Test Module:

This module allows students to take quizzes and evaluate their knowledge. It includes question display, answer selection, automatic evaluation, and result generation. The results are stored in the database for performance tracking.

6. Trainer Module:

This module provides functionalities for trainers to monitor student progress and performance. Trainers can use this data to provide better guidance and support.

7. Database Design:

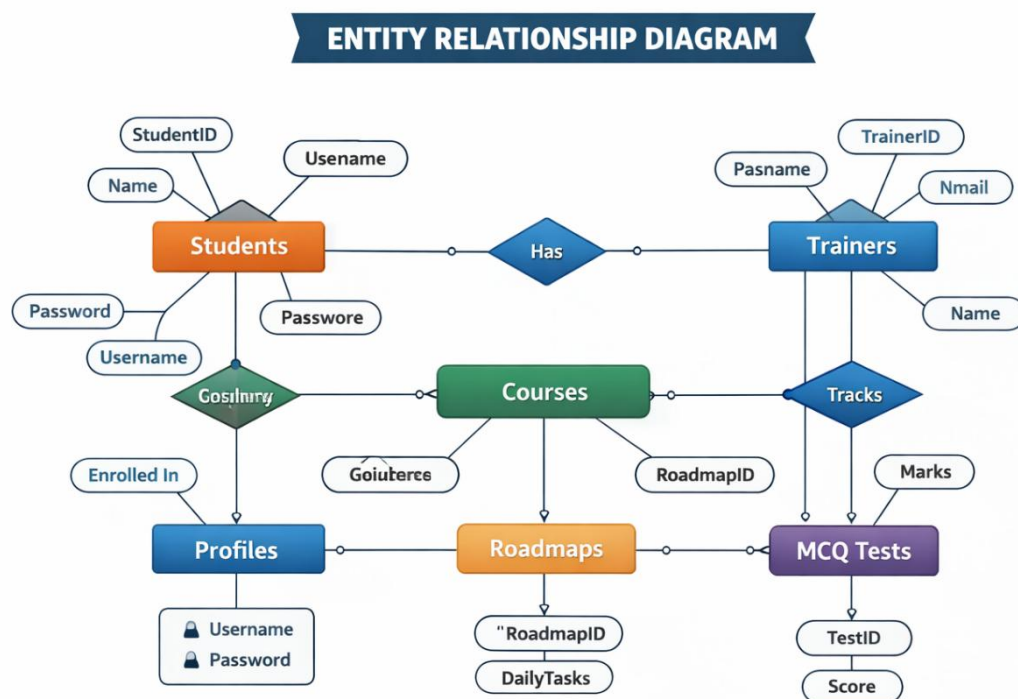
The database is structured into multiple tables such as Users, Courses, Roadmaps, Test Results, and OCR Data. Relationships between tables ensure efficient data storage and retrieval. Proper normalization is applied to reduce redundancy and maintain data integrity.

8. API Design:

The system uses RESTful APIs to connect the frontend and backend. Each module

communicates through specific endpoints, ensuring a clean and organized structure. API testing using Postman ensures that all endpoints function correctly.

The overall system design ensures that the application is scalable, maintainable, and efficient. By separating concerns into different modules and using modern web technologies, the system provides a robust solution for personalized learning and performance tracking.



3.2.1 Entity Relationship Diagram

3.2.2 WORKFLOW IMPLEMENTATION

INTRODUCTION

The Workflow Implementation of the Training Recommendation System defines the step-by-step process through which the system operates, from user interaction to data

processing and result generation. It ensures smooth coordination between different modules and provides a seamless experience for both students and trainers.

The workflow begins with the user authentication process. When a user accesses the system, they are required to either register or log in. During registration, user details such as name, email, password, and role (student or trainer) are collected and securely stored in the database using password hashing techniques. During login, the system verifies the credentials, and upon successful authentication, a JWT (JSON Web Token) is generated. This token is used to maintain a secure session and allow access to authorized features within the system.

Once authenticated, the user is redirected to their respective role-based dashboard. If the user is a student, they are taken to the student dashboard, where they can view their current course, course level, and learning goals. Based on this information, the system initiates the roadmap generation process. The backend processes the input data and dynamically generates a personalized daily learning roadmap, which is displayed to the student. This roadmap helps the student follow a structured plan and stay consistent in their learning.

The next step in the workflow involves the OCR module. Students can upload images containing text, such as handwritten notes or printed materials. The system processes the image using Optical Character Recognition techniques to extract the text. The extracted content is then formatted and presented in a readable manner, allowing students to easily review and understand the information.

In parallel, the profile management module allows users to update and manage their personal information. Students and trainers can modify their details, ensuring that their profiles remain up to date.

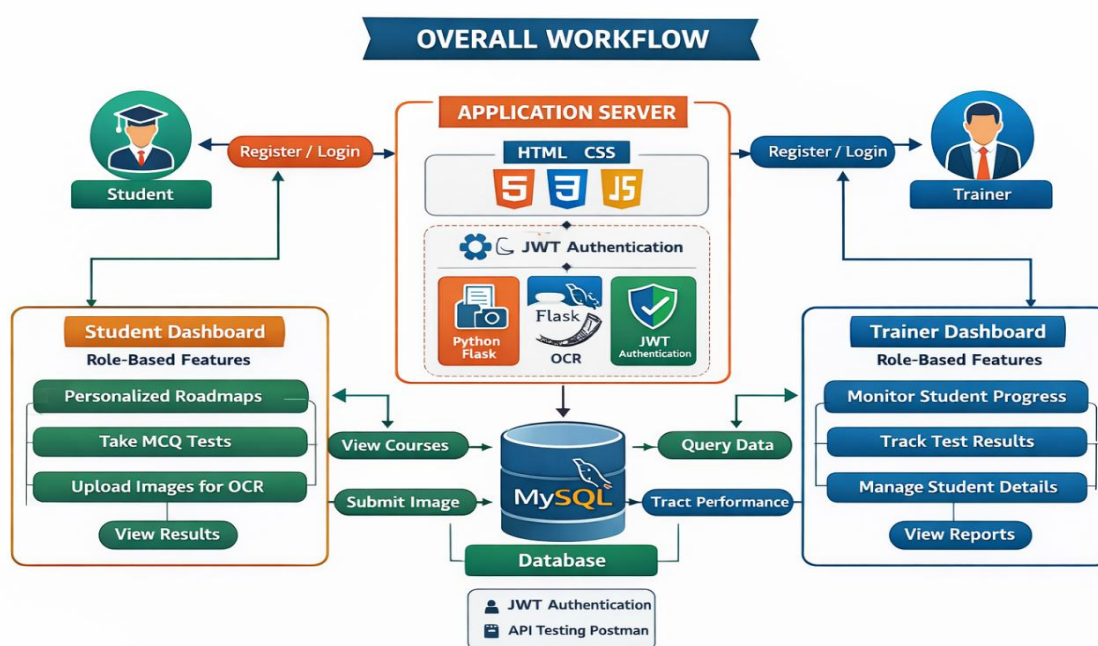
For trainers, the workflow includes accessing the trainer dashboard, where they can monitor student activities. Trainers can view student progress, analyze test results, and track overall performance. The system retrieves data from the database and presents it in an organized format, enabling trainers to make informed decisions and provide appropriate guidance.

Throughout the workflow, the system ensures efficient communication between frontend and backend through RESTful APIs. All user actions, such as login, roadmap generation,

test submission, and OCR processing, are handled via API requests and responses. These APIs are tested using Postman to ensure accuracy and reliability.

Finally, all data generated during the workflow, including user information, roadmap details, OCR outputs, and test results, are stored in the MySQL database. This ensures data consistency, easy retrieval, and scalability of the system.

In conclusion, the workflow implementation of the Training Recommendation System is designed to be systematic, secure, and efficient. It integrates multiple modules seamlessly to provide a smooth user experience, enabling personalized learning for students and effective monitoring for trainers.



3.2.2 Overall Workflow Diagram

CHAPTER – IV

PROPOSED WORK MODULES

4.1 PROPOSED WORK

The proposed work of the Training Recommendation System focuses on designing and developing a smart, secure, and user-friendly web-based platform that provides learning guidance and efficient performance tracking. The system aims to overcome the limitations of traditional learning methods by integrating modern technologies, automation, and data-driven approaches to enhance the overall educational experience.

The core idea of the proposed system is to implement a role-based architecture that supports two primary users: students and trainers. Each user is provided with a dedicated dashboard that offers functionalities tailored to their needs. The system ensures secure access through a login and registration module, where user credentials are protected using password hashing techniques and JWT-based authentication. This ensures that only authorized users can access system features, maintaining data privacy and security.

For students, the proposed system introduces a personalized learning environment. After logging in, students can view their current course details, learning level, and goals. Based on this information, the system dynamically generates a daily learning roadmap. This roadmap serves as a structured plan that guides students step-by-step, helping them stay organized and focused. The roadmap generation process is designed to adapt to individual learning requirements, making the system flexible and efficient.

Another significant component of the proposed work is the integration of an Optical Character Recognition (OCR) module. This feature allows students to upload images containing text, such as handwritten notes or printed documents. The system processes these images and extracts textual content, presenting it in a clear and readable format. This functionality enhances accessibility and helps students easily convert physical study materials into digital form for better understanding and revision.

The system also includes a self-evaluation mechanism through an MCQ test module. Students can attempt quizzes related to their courses and receive instant feedback on their performance. The system evaluates answers automatically, calculates scores, and stores the

results in the database. This enables students to continuously assess their knowledge and identify areas for improvement, promoting active learning.

4.1.1 USER AUTHENTICATION AND ROLE-BASED ACCESS

User Authentication and Role-Based Access are fundamental components of the Training Recommendation System, ensuring that the application is secure, reliable, and accessible only to authorized users. These mechanisms help protect sensitive user data while providing customized functionalities based on user roles.

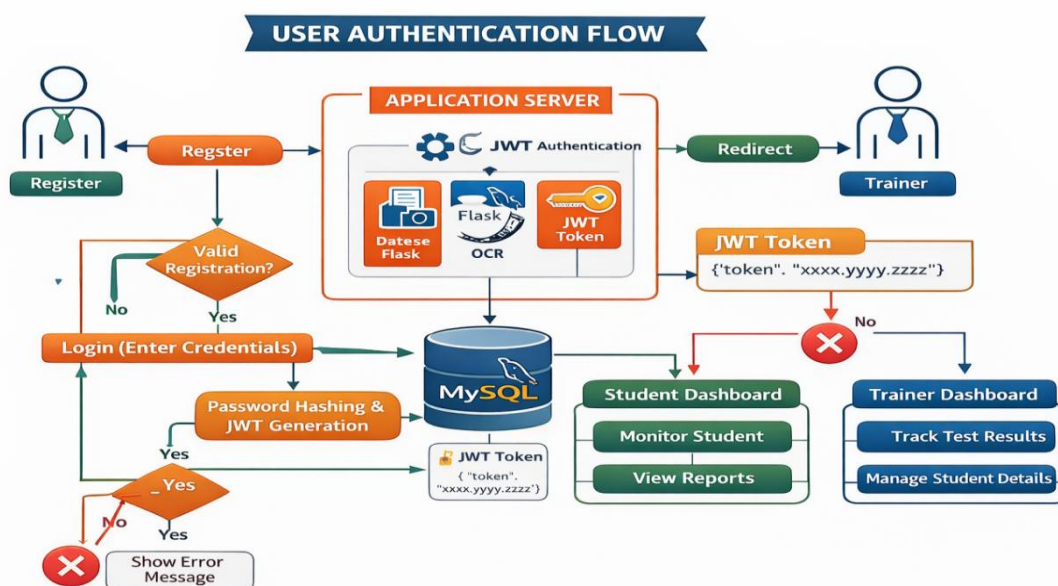
The system begins with a user registration process, where new users are required to provide essential details such as name, email, password, and role (student or trainer). During this process, the system ensures that all inputs are validated to prevent errors and duplicate entries. The passwords entered by users are not stored in plain text; instead, they are securely encrypted using hashing techniques. This ensures that even if the database is compromised, user credentials remain protected.

Once registered, users can access the system through the login module. During login, the entered credentials are verified against the stored data in the database. If the credentials are valid, the system generates a JSON Web Token (JWT), which acts as a secure session identifier. This token is sent to the client side and is used for authenticating subsequent requests made by the user. The use of JWT eliminates the need for server-side session storage and enhances the scalability of the system.

The system implements role-based access control (RBAC) to differentiate between student and trainer functionalities. Each user is assigned a specific role during registration, and access permissions are defined accordingly. After successful login, users are redirected to their respective dashboards based on their roles.

Students: The system provides access to features such as viewing course details, setting learning goals, generating personalized roadmaps, uploading images for OCR processing, attempting MCQ tests, and managing their profiles. Students can only access their own data, ensuring privacy and data security.

Trainers: The system provides access to monitoring and evaluation features. Trainers can view student progress, track test scores, and analyze overall performance. However, they do not have access to modify student credentials or sensitive personal information, maintaining strict access boundaries.



4.1.1 User Authentication Flow

4.1.2 STUDENT MODULE

The Student Module is one of the core components of the Training Recommendation System, designed to provide a personalized and interactive learning experience for students. This module enables students to access learning resources, track their progress, and improve their knowledge through structured guidance and self-assessment tools.

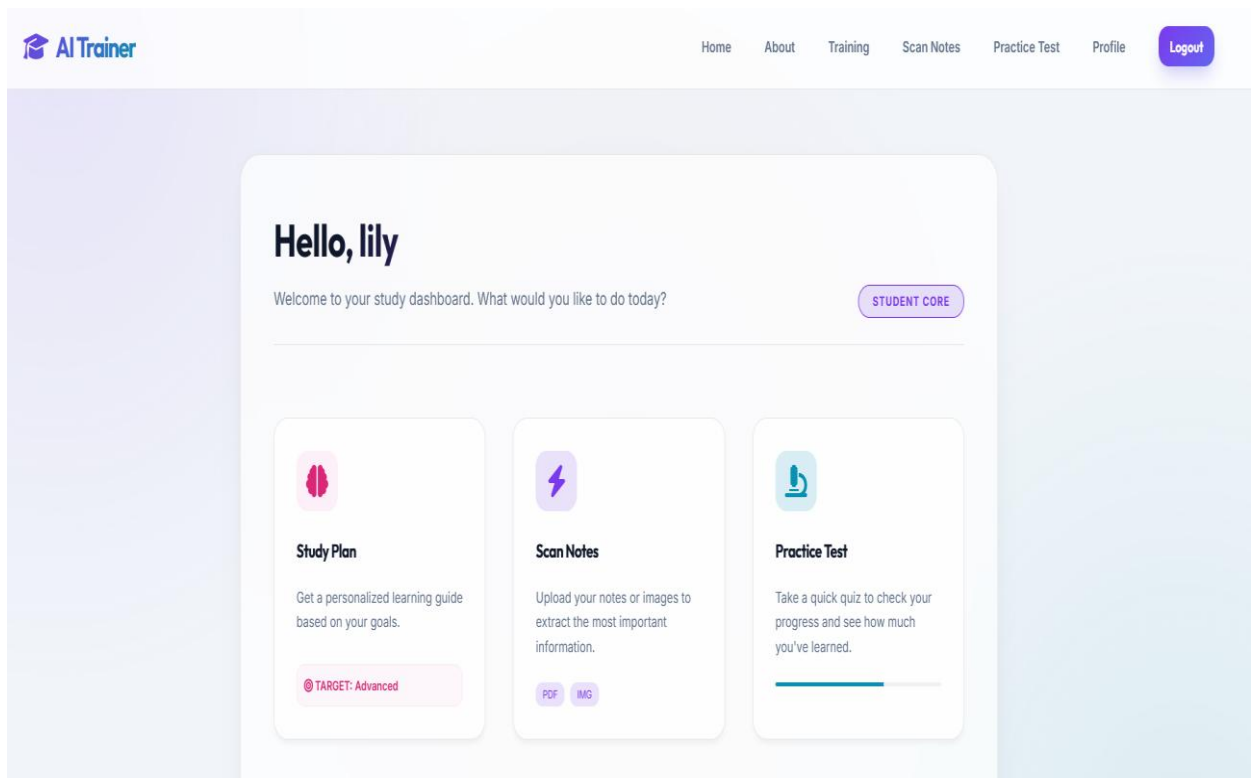
One of the key features of the Student Module is the personalized roadmap generation. Based on the student's selected course and goals, the system dynamically creates a day-by-day learning plan. This roadmap provides clear guidance on what topics to study and what tasks to complete, ensuring a structured and efficient learning process. It helps students manage their time effectively and maintain consistency in their studies.

The module also includes an OCR (Optical Character Recognition) feature, which enhances learning by allowing students to upload images containing text. These images may include handwritten notes, textbook pages, or printed materials. The system processes the uploaded

image, extracts the text, and displays it in a clean and readable format. This makes it easier for students to review and understand the content without manually typing or rewriting it.

Another important functionality is the MCQ test module, which supports self-evaluation. Students can attempt quizzes related to their courses and receive immediate feedback. The system automatically evaluates the answers, calculates scores, and displays the results. This helps students identify their strengths and weaknesses, enabling them to focus on areas that require improvement.

The Student Module also provides a result and performance tracking feature. Students can view their previous test results and monitor their progress over time. This continuous feedback mechanism motivates students to improve and achieve better outcomes.



4.1.2 Student Dashboard

4.1.3 TRAINER MODULE

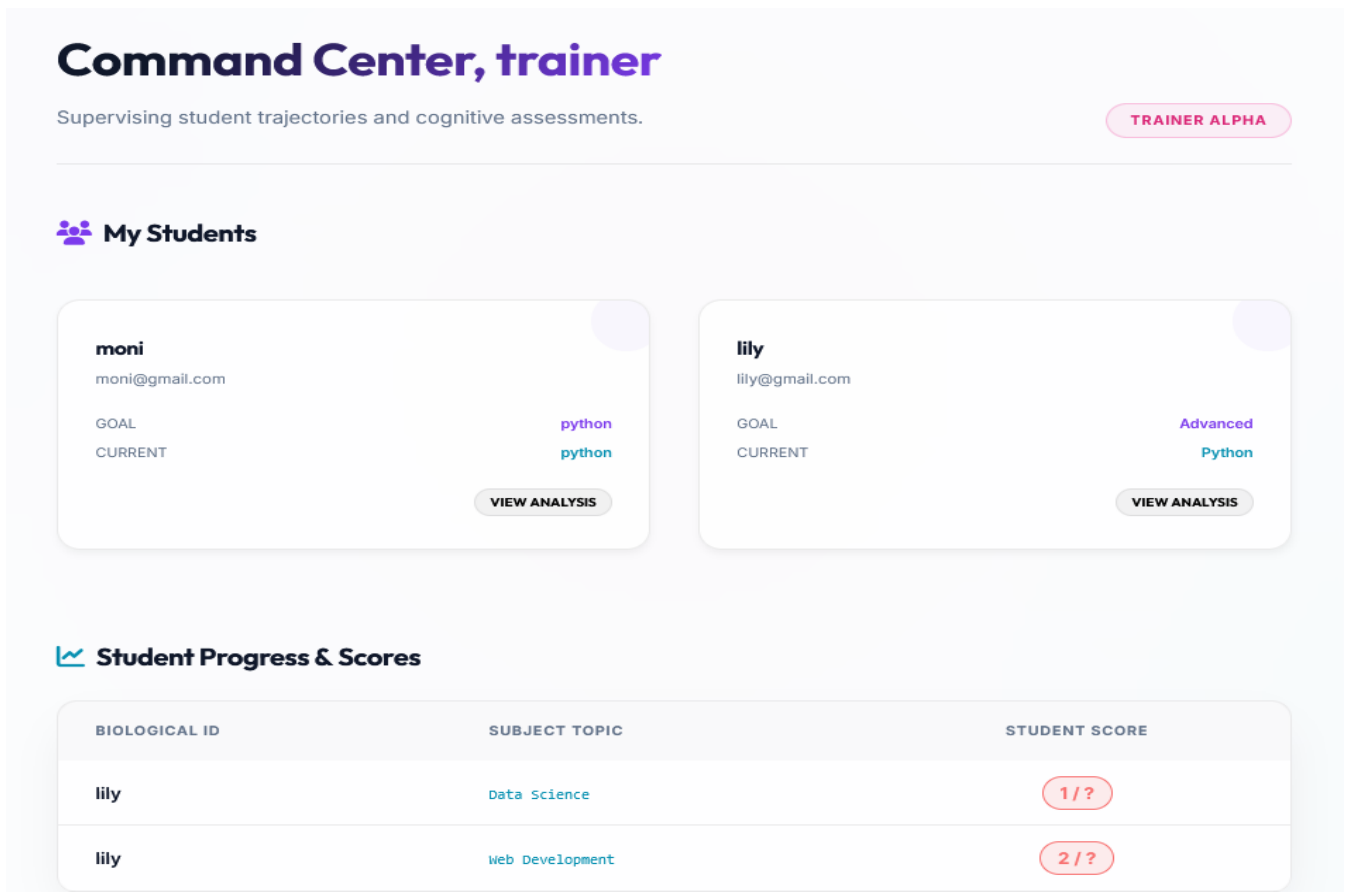
The Trainer Module is a vital component of the Training Recommendation System, designed to help trainers efficiently monitor, evaluate, and manage student learning activities. This module provides a centralized platform where trainers can access student data, track progress, and ensure that learning objectives are being achieved effectively.

The module begins with secure authentication and access control, where trainers log in using their credentials. Once authenticated, they are redirected to the trainer dashboard, which serves as the main interface for all trainer-related functionalities. This dashboard is designed to provide a clear overview of student performance and system activities.

One of the primary features of the Trainer Module is student progress monitoring. Trainers can view details about each student, including their enrolled courses, current learning levels, and progress status. This allows trainers to understand how well students are following their personalized learning roadmaps and identify those who may need additional support.

The module also includes a test performance tracking system. Trainers can access the results of MCQ tests attempted by students, including scores and performance trends. This helps in evaluating the effectiveness of the learning process and identifying strengths and weaknesses among students. By analyzing these results, trainers can provide better feedback and guidance.

Another important functionality is student data management. Trainers can view and manage student-related information in an organized manner. While trainers have access to academic and performance-related data, the system ensures that sensitive information such as login credentials remains secure and inaccessible, maintaining privacy and data integrity.



4.1.3 Trainer Dashboard

4.1.4 MCQ ASSESSMENT MODULE

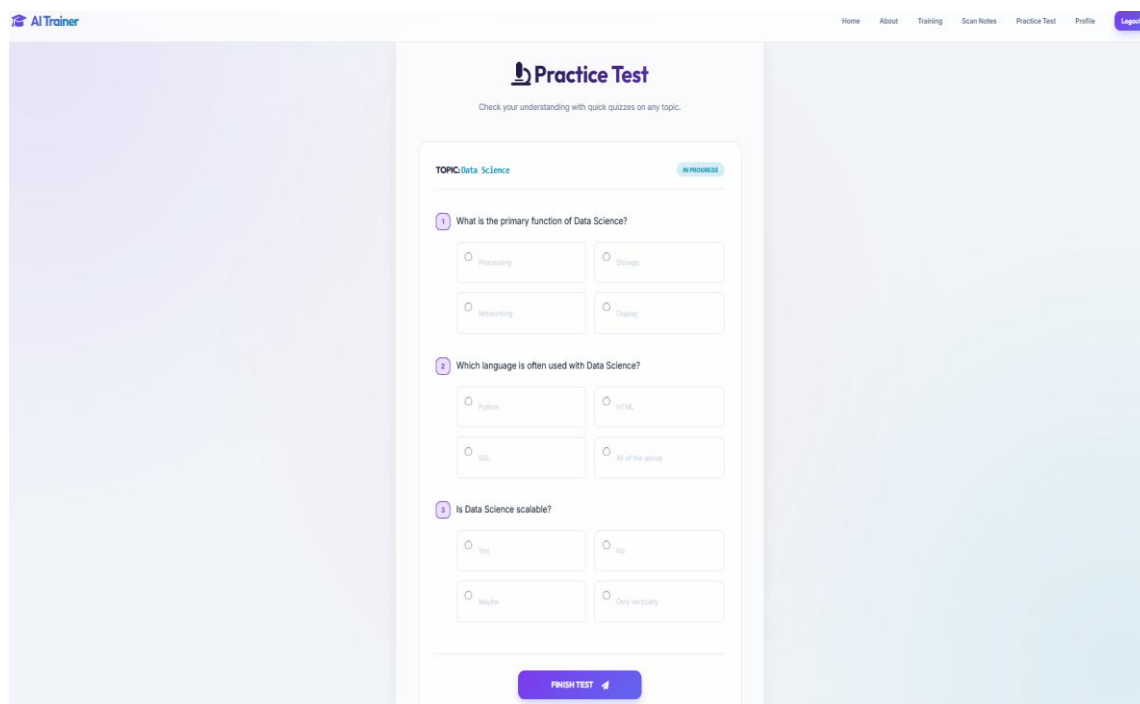
The MCQ Assessment Module is an essential component of the Training Recommendation System, designed to evaluate student knowledge, reinforce learning, and provide instant feedback. This module enables students to test their understanding of various topics through structured multiple-choice questions and helps trainers monitor academic performance effectively.

The module allows students to access and attempt quizzes based on their selected courses or learning topics. Each quiz consists of a set of multiple-choice questions with predefined options, where only one correct answer is expected. The questions are stored in the database and are dynamically retrieved and displayed to the student through the user interface.

When a student starts a test, the system presents questions in an organized format, ensuring a smooth and user-friendly experience. Students can select their answers and submit the test upon completion. The system then performs automatic evaluation, comparing the selected answers with the correct answers stored in the database.

One of the key features of this module is real-time result generation. Immediately after submission, the system calculates the total score and displays the results to the student. This instant feedback helps students quickly understand their performance and identify areas where improvement is needed.

For trainers, the MCQ Assessment Module provides tools for analyzing student performance. Trainers can access test results, evaluate overall class performance, and identify trends such as commonly incorrect questions or weak subject areas. This information helps trainers adjust teaching strategies and provide targeted guidance to students.



The screenshot displays the 'Practice Test' interface on the 'AI Trainer' platform. The top navigation bar includes links for Home, About, Training, Scan Notes, Practice Test, Profile, and a Logout button. The main content area is titled 'Practice Test' with a subtitle 'Check your understanding with quick quizzes on any topic.' Below this, a section for 'TOPIC: Data Science' is shown with a 'IN PROGRESS' status. The test consists of three multiple-choice questions:

1. What is the primary function of Data Science?
☐ Processing ☐ Storage
☐ Networking ☐ Display
2. Which language is often used with Data Science?
☐ Python ☐ HTML
☐ SQL ☐ All of the above
3. Is Data Science scalable?
☐ Yes ☐ No
☐ Maybe ☐ Only vertically

A 'FINISH TEST' button is located at the bottom of the question list.

4.1.4 MCQ Assessment

4.1.5 OCR PROCESSING MODULE

The OCR (Optical Character Recognition) Processing Module is an advanced feature of the Training Recommendation System that enhances the learning experience by converting image-based text into editable and readable digital content. This module is especially useful for students who rely on handwritten notes, printed materials, or scanned documents as part of their study process.

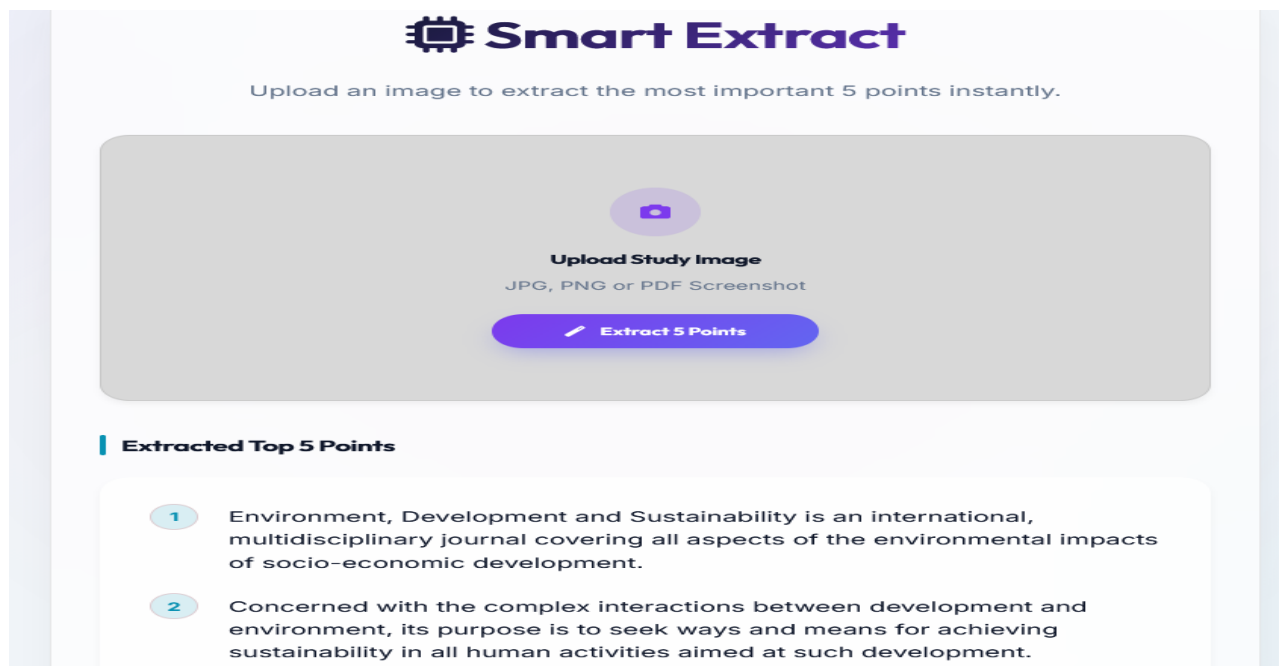
The workflow of the OCR module begins with the image upload functionality, where students can submit images containing textual information. These images may include photographs of handwritten notes, textbook pages, or printed documents. The system accepts various image formats and ensures proper handling of the uploaded files.

Once the image is uploaded, it is processed by the backend using OCR techniques implemented through Python-based libraries integrated with the Flask framework. The system performs image preprocessing steps such as noise reduction, grayscale conversion, and contrast enhancement to improve the accuracy of text extraction. These preprocessing techniques help in handling different image qualities and lighting conditions.

After preprocessing, the system applies text recognition algorithms to extract the textual content from the image. The extracted text is then converted into a machine-readable format and sent back to the frontend through API responses. This process is performed efficiently to provide quick results to the user.

The extracted content is displayed in a clean and structured format, making it easier for students to read and understand. The system may also highlight key points or organize the text into paragraphs for better readability. This eliminates the need for manual typing and allows students to quickly digitize their study materials.

The OCR module also supports data storage and retrieval, where extracted text can be saved in the MySQL database for future reference. This enables students to revisit previously processed content without needing to upload the same image again.



4.1.5 OCR Processing

4.1.6 RECOMMENDATION ENGINE MODULE

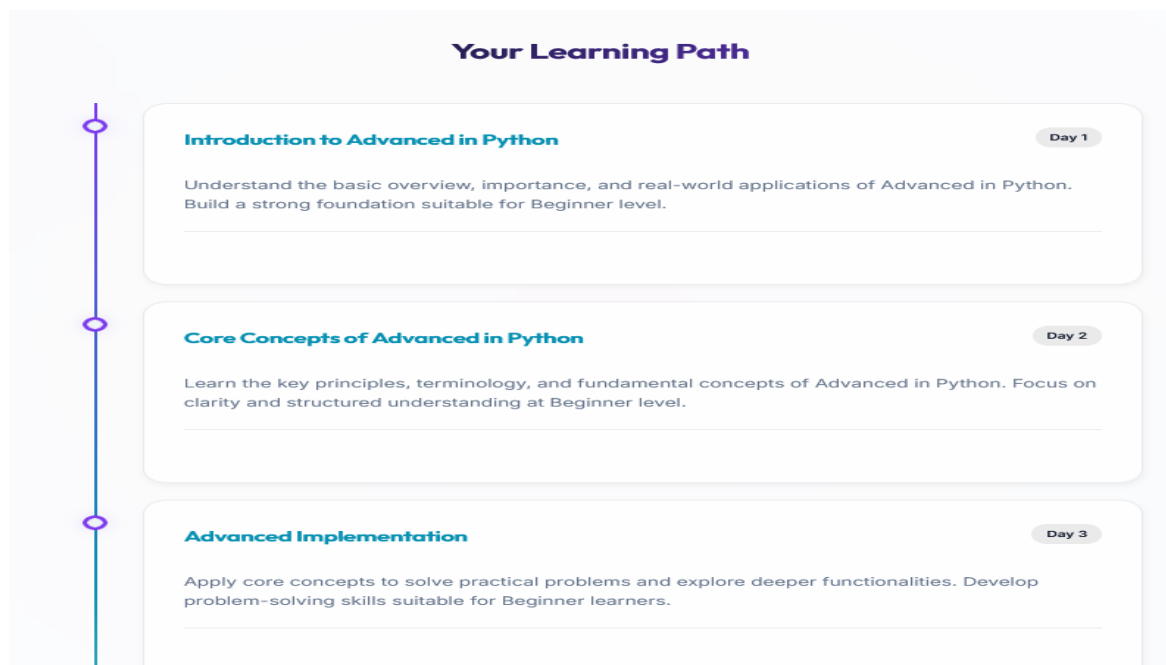
The Recommendation Engine Module is a core component of the Training Recommendation System, responsible for generating personalized learning paths and guiding students toward achieving their educational goals efficiently. This module uses user data, learning preferences, and performance metrics to provide tailored recommendations that enhance the overall learning experience.

The process begins with the collection of student data, which includes information such as selected course, current learning level, learning goals, and past performance (e.g., MCQ test results). This data is stored in the MySQL database and serves as the foundation for generating meaningful recommendations.

Based on this data, the system applies a rule-based and logic-driven approach to analyze the student's current status and requirements. For example, if a student is at a beginner level, the system recommends fundamental topics and gradually progresses to advanced concepts. Similarly, if a student performs poorly in certain areas based on test results, the system identifies those weak areas and suggests additional practice or revision topics.

One of the key outputs of this module is the personalized daily learning roadmap. The system dynamically generates a structured plan that includes tasks, topics, and activities to be completed each day. This roadmap helps students follow a disciplined learning schedule and ensures consistent progress toward their goals.

The Recommendation Engine also supports adaptive learning, where recommendations are continuously updated based on user interaction and performance. For instance, if a student improves in a particular topic, the system adjusts future recommendations to include more advanced content. This adaptability makes the learning process more effective and responsive.



4.1.6 Training Recommendation

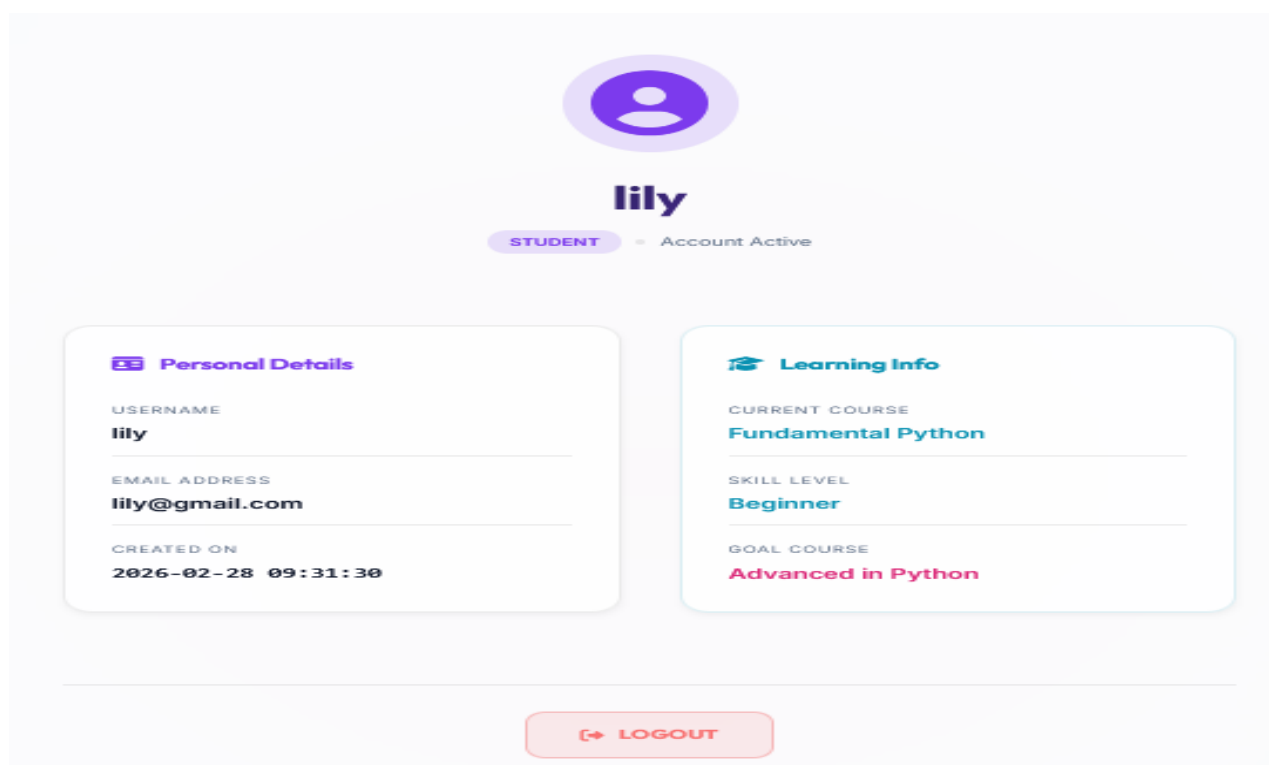
4.1.7 DASHBOARD AND PROFILE MANAGEMENT

The Dashboard and Profile Management module is an essential part of the Training Recommendation System, providing users with a centralized interface to access features, monitor activities, and manage personal information efficiently. This module enhances user experience by offering a well-organized and interactive environment tailored to the specific roles of students and trainers.

The system includes role-based dashboards, which are displayed after successful user authentication. Based on the assigned role, users are redirected to either the student dashboard or the trainer dashboard. Each dashboard is designed with relevant features and functionalities to meet the specific needs of the user.

Student dashboard : Serves as the primary interface for learners. It displays important details such as the current course, course level, and learning goals. It also provides access to key features like personalized roadmap generation, MCQ tests, OCR image upload, and performance tracking. The dashboard is designed to be intuitive and user-friendly, allowing students to easily navigate through different sections and monitor their learning progress in real time.

Trainer dashboard : Focuses on monitoring and evaluation. Trainers can view student progress, analyze test results, and track overall performance. The dashboard presents data in a structured format, enabling trainers to quickly identify trends and provide appropriate guidance. It acts as a control panel for managing student-related activities and ensuring effective learning outcomes.



4.1.7 Dashboard and Profile Management

CHAPTER – V SYSTEM IMPLEMENTATION

5.1 FRONTEND IMPLEMENTATION (HTML,CSS,JS)

The Frontend Implementation of the Training Recommendation System focuses on designing an interactive, responsive, and user-friendly interface that enables seamless interaction between users and the system. The frontend is developed using HTML, CSS, and JavaScript, which together provide structure, styling, and dynamic behavior to the application.

HTML (HyperText Markup Language) : It is used to structure the content of the web pages. It defines elements such as forms, buttons, input fields, and layout sections for different modules like login, dashboard, and test pages. Proper semantic structuring ensures better readability and maintainability of the code.

CSS (Cascading Style Sheets) : It is used to enhance the visual appearance of the application. It provides styling for layouts, colors, fonts, and responsiveness. The system follows a consistent design theme to maintain a professional look across all pages. Responsive design techniques are implemented to ensure compatibility with different screen sizes, including desktops, tablets, and mobile devices.

JavaScript : It plays a crucial role in adding interactivity and dynamic functionality to the frontend. It is used for handling user inputs, validating forms, managing events, and making API calls to the backend. For example, when a user submits a login form, JavaScript captures the input data and sends it to the backend for authentication. Similarly, it handles dynamic content updates such as displaying roadmap data, showing quiz results, and rendering OCR outputs.

The frontend also includes form validation mechanisms to ensure that users enter correct and complete information. Validation checks are performed for fields such as email format,

password strength, and required inputs. This reduces errors and improves data quality before sending requests to the backend.

The system also implements role-based UI rendering, where different interfaces are displayed based on the user's role (student or trainer). This ensures that users only see features relevant to them, improving usability and security

5.2 BACKEND IMPLEMENTATION (PYTHON – FLASK FRAMEWORK)

The Backend Implementation of the Training Recommendation System is responsible for handling the core functionality, business logic, data processing, and communication between the frontend and the database. It ensures that all user requests are processed efficiently, securely, and accurately. The backend of the system is developed using Python Flask, a lightweight and flexible web framework that supports rapid development and easy integration of various modules.

One of the primary responsibilities of the backend is user authentication and authorization. The system verifies user credentials during login and generates a JSON Web Token (JWT) for authenticated sessions. This token is used to validate user identity for subsequent requests. Passwords are securely stored using hashing techniques, ensuring that sensitive information is protected. The backend also enforces role-based access control, ensuring that students and trainers can only access features relevant to their roles.

The backend includes a Recommendation Engine, which processes user data such as course details, learning goals, and performance metrics to generate personalized daily roadmaps. This logic is implemented using rule-based algorithms that adapt to individual user needs, providing a customized learning experience.

Another important component is the OCR Processing Module, which handles image uploads and extracts text using OCR libraries. The backend processes the uploaded images, applies preprocessing techniques, and returns the extracted text to the frontend in a structured format.

5.3 DATABASE DESIGN (MYSQL)

The Database Design of the Training Recommendation System plays a crucial role in managing and organizing data efficiently. It ensures that all user information, course details, learning progress, and system-generated outputs are stored in a structured and consistent manner. The system uses MySQL as the database management system due to its reliability, scalability, and support for relational data models.

Key features include:

- **Database Tables:**
 - **Users:** Stores details of Students, Trainers, and Admins.
 - **Courses/Trainings:** Contains all available trainings with details like duration, category, and prerequisites.
 - **Recommendations:** Stores recommended courses for each student based on analysis.
 - **Assessments:** Stores MCQ or assignment results to evaluate skills.
 - **Feedback & Ratings:** Stores user feedback on courses for further refinement of recommendations.
- **Relationships:** Primary and foreign keys ensure accurate mapping between users, courses, and recommendations.
- **CRUD Operations:** The backend performs create, read, update, and delete operations for courses, user profiles, and assessments efficiently.
- **Indexing:** Frequently accessed fields like user IDs, course IDs, and recommendation scores are indexed to enhance query performance.
- **Security & Data Integrity:** Role-based access, data validation, and password encryption ensure safe and reliable storage of sensitive information.

CHAPTER – VI RESULTS AND DISCUSSION

6.1 SYSTEM PERFORMANCE EVALUATION

6.1.1 USER AUTHENTICATION PERFORMANCE

The performance of the User Authentication module in the Training Recommendation System is a critical factor in ensuring a smooth, secure, and efficient user experience. This module is designed to handle user login and registration requests quickly while high security standards through the use of password hashing and JWT-based authentication.

During the authentication process, user credentials are validated against the database with optimized queries, ensuring minimal response time even when multiple users access the system simultaneously. The use of hashed passwords enhances security without significantly impacting performance, as modern hashing algorithms are both secure and computationally efficient.

Additionally, the implementation of JSON Web Tokens (JWT) improves performance by enabling stateless authentication, eliminating the need for repeated database lookups for session validation. Once a token is generated, it is used for subsequent requests, reducing server load and improving response speed. The system also includes proper error handling and validation mechanisms, ensuring that invalid login attempts are processed efficiently without affecting overall performance.

6.1.2 MCQ EVALUATION ACCURACY

The MCQ Evaluation Module in the Training Recommendation System ensures high accuracy in assessing student performance. The system automatically compares the answers selected by students with the correct answers stored in the database. This eliminates human error and ensures consistency in evaluation. The scoring process is instant and reliable, providing immediate feedback to users.

Additionally, the system maintains records of past performances, allowing students and trainers to track progress over time. The structured design of the question bank and validation logic ensures that the evaluation process remains precise, efficient, and scalable for multiple users.

6.1.3 OCR PROCESSING PERFORMANCE

The OCR Processing Module is designed to efficiently extract text from images uploaded by users. The system utilizes OCR libraries to convert image-based content into readable and editable text. It performs well for clear and high-quality images, providing accurate text extraction within a short processing time.

Basic image preprocessing techniques such as resizing and noise reduction further enhance accuracy. The extracted content is displayed in a structured format, improving readability and usability for students. Overall, the module balances speed and accuracy, making it a valuable feature for digitizing study materials.

6.1.4 RECOMMENDATION SYSTEM EFFECTIVENESS

The Recommendation System is a core component that enhances personalized learning. It analyzes user inputs such as selected course, current knowledge level, and learning goals to generate customized daily roadmaps. The system effectively guides students by breaking down complex learning paths into manageable tasks.

It adapts to user progress and ensures that recommendations remain relevant and goal-oriented. This improves student engagement, motivation, and overall learning outcomes, making the system highly effective in supporting self-paced education.

6.1.5 DATABASE PERFORMANCE (MYSQL)

The database performance of the system is optimized using MySQL, ensuring efficient data storage and retrieval. Proper normalization reduces redundancy, while indexing improves query execution speed. The system can handle multiple user requests simultaneously without significant delays.

All operations such as login validation, test result storage, roadmap updates, and OCR data handling are executed quickly and reliably. The use of structured queries and well-defined relationships ensures data integrity and smooth backend operations, even as the system scales.

CHAPTER – VII CONCLUSION

7.1 CONCLUSION

The Training Recommendation System successfully achieves its objective of providing a structured, personalized, and efficient learning platform for students while enabling trainers to monitor and guide progress effectively. By integrating modern web technologies such as HTML, CSS, JavaScript, Python Flask, and MySQL, the system ensures smooth interaction between the frontend, backend, and database. The implementation of secure authentication mechanisms using password hashing and JWT enhances system reliability and protects user data.

The system effectively combines multiple modules, including user authentication, personalized roadmap generation, OCR-based text extraction, and MCQ-based self-assessment, to create a comprehensive learning environment. Each module works cohesively to improve the overall learning experience by making it more interactive, organized, and adaptive to individual user needs. The recommendation engine plays a key role in guiding students through a clear learning path, while the OCR and MCQ modules add value by supporting content understanding and self-evaluation.

From a performance perspective, the system demonstrates efficiency, accuracy, and scalability. It minimizes manual effort, reduces errors, and provides real-time feedback, which is essential in modern educational systems. The role-based design ensures that both students and trainers have access to relevant features, improving usability and system effectiveness.

7.2 FUTURE ENHANCEMENTS

The Training Recommendation System has demonstrated robust performance, but there is potential for further improvements to make it more intelligent, interactive, and scalable. The following enhancements can be considered for future versions:

1. **AI-Based Personalized Learning Paths:**

- Integrate machine learning algorithms to dynamically suggest training programs based on user performance, learning preferences, and career goals.
- Use predictive analytics to identify skill gaps and recommend appropriate courses.

2. **Advanced Analytics and Reporting:**

- Implement dashboards with visual analytics to track user progress, training effectiveness, and overall engagement.
- Provide detailed reports for Admins and Trainers to make data-driven decisions.

3. **Gamification and Motivation Features:**

- Introduce gamification elements like badges, leaderboards, and progress tracking to encourage user engagement.
- Provide rewards or certifications for completing training milestones.

4. **Mobile Application Integration:**

- Develop a mobile app version to allow users to access training resources, complete assessments, and receive recommendations on the go.
- Implement push notifications for reminders, updates, and feedback.

5. **Enhanced Security and Privacy:**

- Implement multi-factor authentication (MFA) for added security.
- Ensure compliance with data privacy regulations such as GDPR and local data protection laws.

6. **Integration with External Learning Platforms:**

- Connect with popular e-learning platforms or content providers to expand the range of available training materials.

CHAPTER – VIII REFERENCES

1. Welling, L., & Thomson, L. (2017). Flask Web Development: Developing Web Applications with Python. O'Reilly Media.
2. Duckett, J. (2014). HTML & CSS: Design and Build Websites. Wiley.
3. Freeman, E., & Robson, E. (2014). JavaScript: The Good Parts. O'Reilly Media.
4. MySQL Documentation. (2025). MySQL 8.0 Reference Manual. Retrieved from <https://dev.mysql.com/doc/>
5. Richardson, L., & Ruby, S. (2007). RESTful Web Services. O'Reilly Media.
6. Brownlee, J. (2020). Machine Learning Mastery: How to Build a Recommendation System. Machine Learning Mastery.
7. MongoDB Manual (2025). Data Modeling, Indexing, and Performance. Retrieved from <https://www.mongodb.com/docs/manual/>
8. MDN Web Docs. (2025). HTML, CSS, and JavaScript Guides. Mozilla Developer Network. Retrieved from <https://developer.mozilla.org/>
9. Flask Official Documentation. (2025). Flask Web Framework. Retrieved from <https://flask.palletsprojects.com/>
10. Ricci, F., Rokach, L., Shapira, B., & Kantor, P. B. (2011). Recommender Systems Handbook. Springer.
11. Kotsiantis, S., Zaharakis, I., & Pintelas, P. (2007). Supervised Machine Learning: A Review of Classification Techniques. Emerging Artificial Intelligence Applications in Computer Engineering.

CHAPTER - IX APPENDICES

The development of the Training Recommendation System was carried out individually, where all modules of the system were designed and implemented to ensure smooth functionality, including user management, authentication, MCQ evaluation, OCR processing, task assignment, recommendation system, and database performance optimization.

DEVELOPER : MONIKA N

CONTRIBUTIONS:

- Developed the frontend using HTML, CSS, and JavaScript, creating interactive and responsive user interfaces.
- Designed and implemented user login and registration pages, ensuring a smooth authentication flow.
- Created forms and dashboards for Trainer and Student roles with proper input validation.
- Integrated the frontend with the Python Flask backend using APIs for dynamic data exchange.
- Designed and managed the database using MySQL, including tables for users, tasks, and evaluation results.
- Implemented security measures such as input validation and token-based session management for secure authentication.
- Developed core functionalities including MCQ evaluation, OCR processing, task assignment, and recommendation system.
- Optimized database queries and system performance for efficient data handling.
- Conducted testing for performance, usability, and error handling, ensuring a smooth and reliable user experience.

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